25900520 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMBG203 - FOODBORNE DISEASES, FOOD SANITATION AND FOOD SAFETY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Bacterial food infection occurs when: [U] / [CO1]

   a). Bacterial toxins are ingested

   b). Allergens are ingested

   c). Living bacteria are ingested

   d). Chemicals are ingested

2. Staphylococcal enterotoxin acts on : [U] / [CO2]

   a). Intestine b). Brain

   c). Nerves d). Lungs

3. The primary mode of transmission of Norovirus is: [U] / [CO1]

   a). Faecal-oral route b). Vector-borne

   c). Blood transfusion d). Airborne

4. Which of the following is not a biological hazard in food? [K] / [CO1]

a). Norovirus b). Parasites

c). Pesicides d). Bacteria

5. Aflatoxins are produced mainly by: [U] / [CO2]

a). Aspergillus b). Fusarium

c). Claviceps d). Streptococcus

6. The main target organ affected in aflatoxicoses: [U] / [CO2]

a). Brain b). Liver

c). Eyes d). Lungs

7. The mode of transmission of bacillary dysentery is: [U] / [CO2]

a). Faecal-oral route b). Vector-borne

c). Blood transfusion d). Airborne

8. Which of the following is not a food intoxication? [K] / [CO1]

a). Botulism b). Staphylococcal poisoning

c). Salmonellosis d). Aflatoxicosis

9. The main route of transmission of salmonellosis is: [U] / [CO1]

a). Faecal-oral route b). Vector-borne

c). Air borne d). Blood transfusion

10. GMP stands for _____. [K] / [CO4]

a). Good Manufacturing Practices

b). General Manufacturing Procedure

c). Government Managed Production

d). Global Market Policy

[1x10 = 10]

Part B

Fill in the blanks questions

Answer all questions

Each question carries **1** marks

11. Patulin is produced by _____. [U] / [CO2]

12. Rotavirus infection is the most common cause of severe _____ in infants and young children worldwide. [K] / [CO2]

13. _____ are the organizations that participate in the Codex Alimentarius Commission's work without having voting rights. [U] / [CO4]

14. Monitoring is a planned sequence of observations or measurements of control parameters to assess whether a_____ is under control. [U] / [CO4]

15. QA is a _____ approach aimed to prevent problems from occurring during production. [U] / [CO4]

[1x5 = 5]

Part C

True or false questions

Answer all questions

Each question carries **1** marks

16. *Campylobacter jejuni* is one of the most common causes of bacterial gastroenteritis worldwide. [U] / [CO2]

17. Steam sanitation is ineffective in killing bacterial spores. [K] / [CO3]

18. The ‘temperature danger zone’ in food storage is between 0 degree Celsius and 10 degree Celsius. [K] / [CO4]

19. Trickling filter a method of primary wastewater treatment. [K] / [CO4]

20. Employee training is not required in pest management programs. [K] / [CO4]

[1x5 = 5]

Part D

One word answer questions

Answer any **10** questions

Each question carries **1** marks

21. What is mycotoxicosis? [K] / [CO1]

22. Write the expanded form of the abbreviation FSSAI. [A] / [CO4]

23. Why is segregation important in solid waste management? [K] / [CO4]

24. Which ministry controls AGMARK certification? [K] / [CO4]

25. What is chemical hazard? [K] / [CO1]

26. Select the transmission method of *Ecoli* diarrhoea. [U] / [CO2]

27. How is Hepatitis E transmitted? [K] / [CO2]

28. Name the pathogen responsible for typhoid fever. [K] / [CO2]

29. At what pH is hypochlorites most effective as a sanitizer? [K] / [CO3]

30. Name the pathogen involved in listeriosis. [K] / [CO2]

31. How does a surfactant aid cleaning [K] / [CO3]

32. Identify the type of hazard can metal shavings cause. [U] / [CO1]

[1x10 = 10]

Part E

Short answer questions

Answer any **4** questions

Each question carries **3** marks

33. How can proper food storage and handling practices prevent food borne illness? [K] / [CO1]

34. What is CIP? List the advantages and disadvantages of CIP system. [K] / [CO3]

35. Discuss the treatment and supportive management of Polio [U] / [CO2]

36. What are the causes and symptoms of Cholera? [K] / [CO2]

37. Describe the role of *Entamoeba histolytica* in causing food borne illness. [U] / [CO2]

38. Identify and describe the benefits organizations can gain by implementing ISO 22000:2018 certification. [U] / [CO4]

[3x4 = 12]

Part F

Long answer questions

Answer any **1** questions

Each question carries **8** marks

39. Explain botulism with reference to incubation period, clinical features, and methods of prevention. [U] / [CO1]

40. List out the key responsibilities of a Quality Control Officer in a food processing unit. [K] / [CO4]

[8x1 = 8]